Polynomial Multiplication Accelerator for FHE

1 Introduction and Modeling

1.1 Polynomial multiplication

- Considering 2 N-degree polynomials:

$$\begin{cases} A = a_0 x^0 + a_1 x^1 + \ldots + a_{N-1} x^{N-1} \\ B = b_0 x^0 + b_1 x^1 + \ldots + b_{N-1} x^{N-1} \end{cases}$$

- We perform multiplication on polynomial rings:

$$R_q = Z_q[x]/(x^N + 1)$$

- So that we can get the result:

$$C = A \times B = c_0 x^0 + c_1 x^1 + \ldots + c_{N-1} x^{N-1}$$

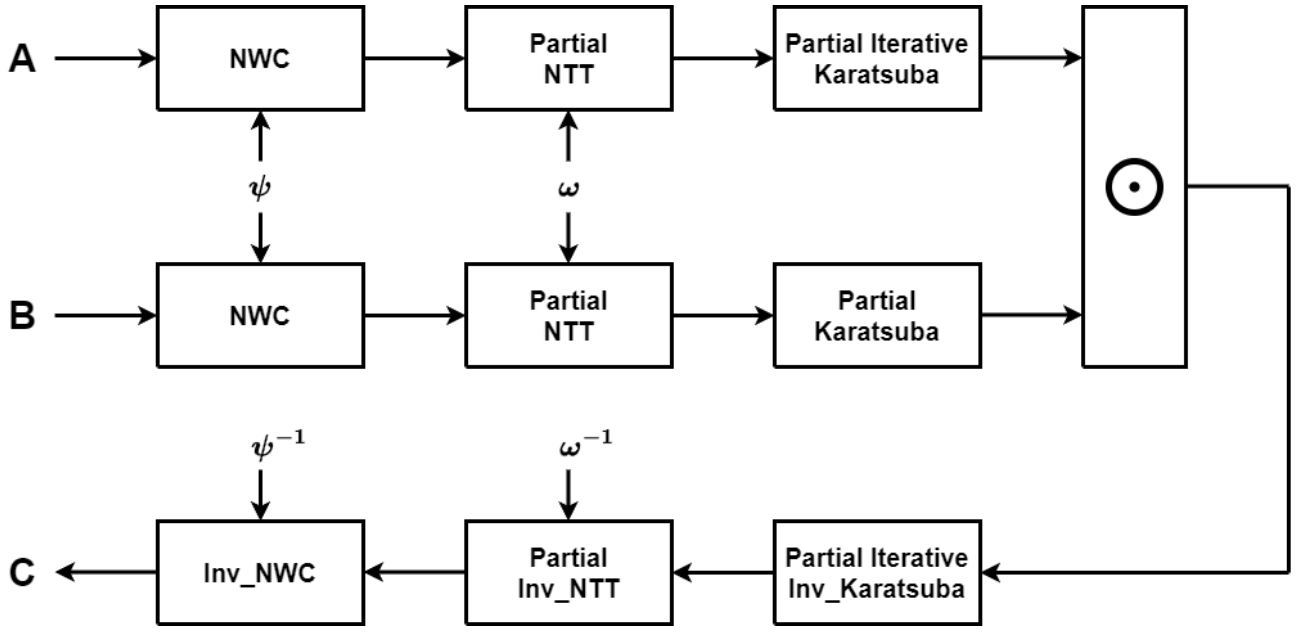
1.2 Solution system

1. The most traditional method for solving such a polynomial multiplication is the naive approach learned in school.
    - In this method, we multiply corresponding elements and sum them according to their respective indices.
    - The computational complexity of this method is $O(N^2)$
2. To accelerate the computation of polynomial multiplication, modern research always empolys the **Number Theoretic Transform (NTT)**, which is an adaption of the FFT that operates in a finite field.
    - Just like FFT, NTT transforms a polynomial into a point-value representation, enabling efficient multiplication in the frequency domain.
    - The computational complexity of this method is $O(NlogN)$
3. The Karatsuba algorithm, originally developed for fast multiplication of large integers, can be adapted for polynomail multiplication.
    - It leverges the divide-and-conquer strategy to reduce the number of multiplications required, replacing some of them with additions and subtractions, which are computationally cheaper.
    - The computational complexity of this method is $O(N^{log_2 3}) \approx O(N^{1.585})$
    - We develop the recursive nature of the Karatsuba algorithm, resulting in the **Iterative Karatsuba Algorithm**, which is highly parallelizable, allowing for optimization on multi-core processors.

- Based upon the 3 methods, the classic solution system of polynomial multiplication is shown below, where $\odot$ represents element-wise multiplication.

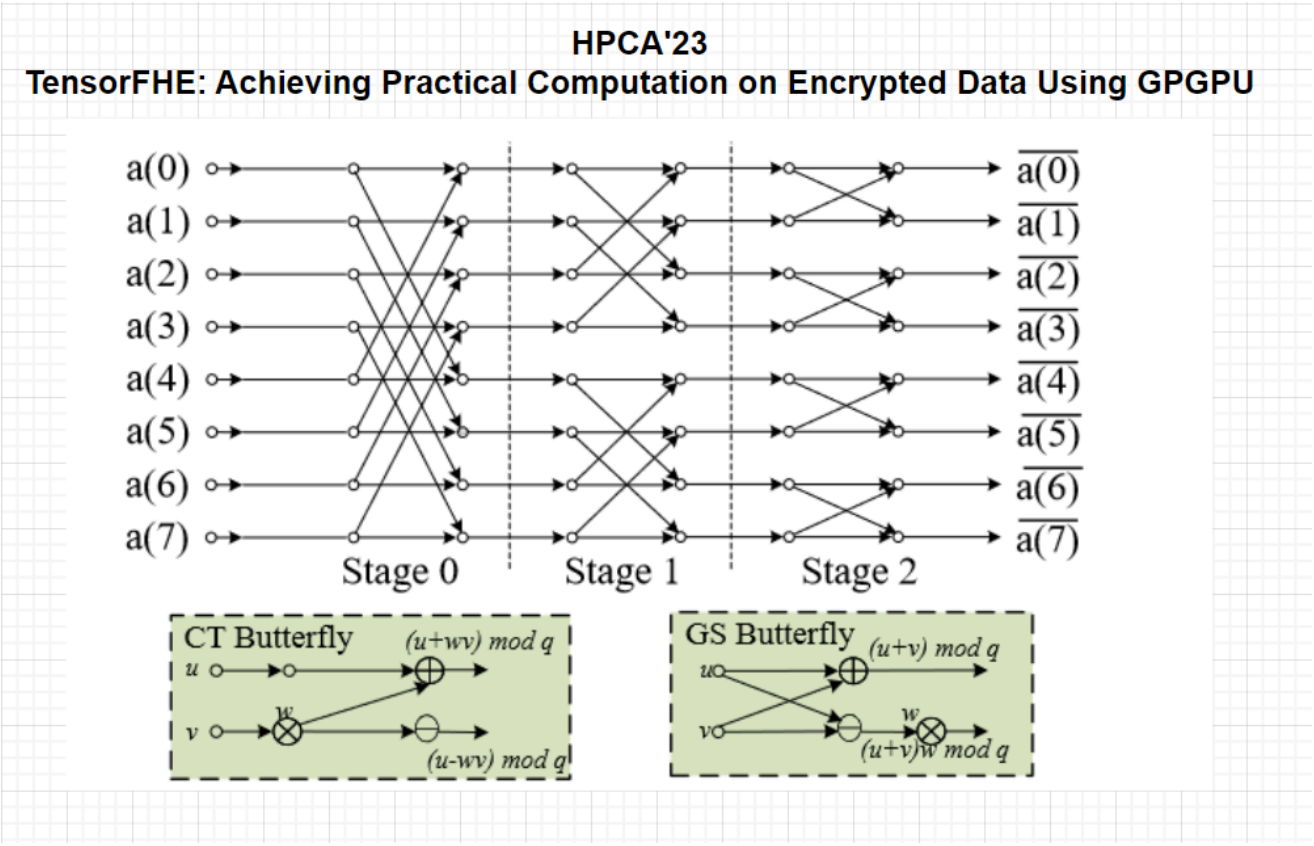
$$INTT(NTT(A) \odot NTT(B))$$

- ***Our IDEA: Partial NTT + Partial Iterative Karatsuba***

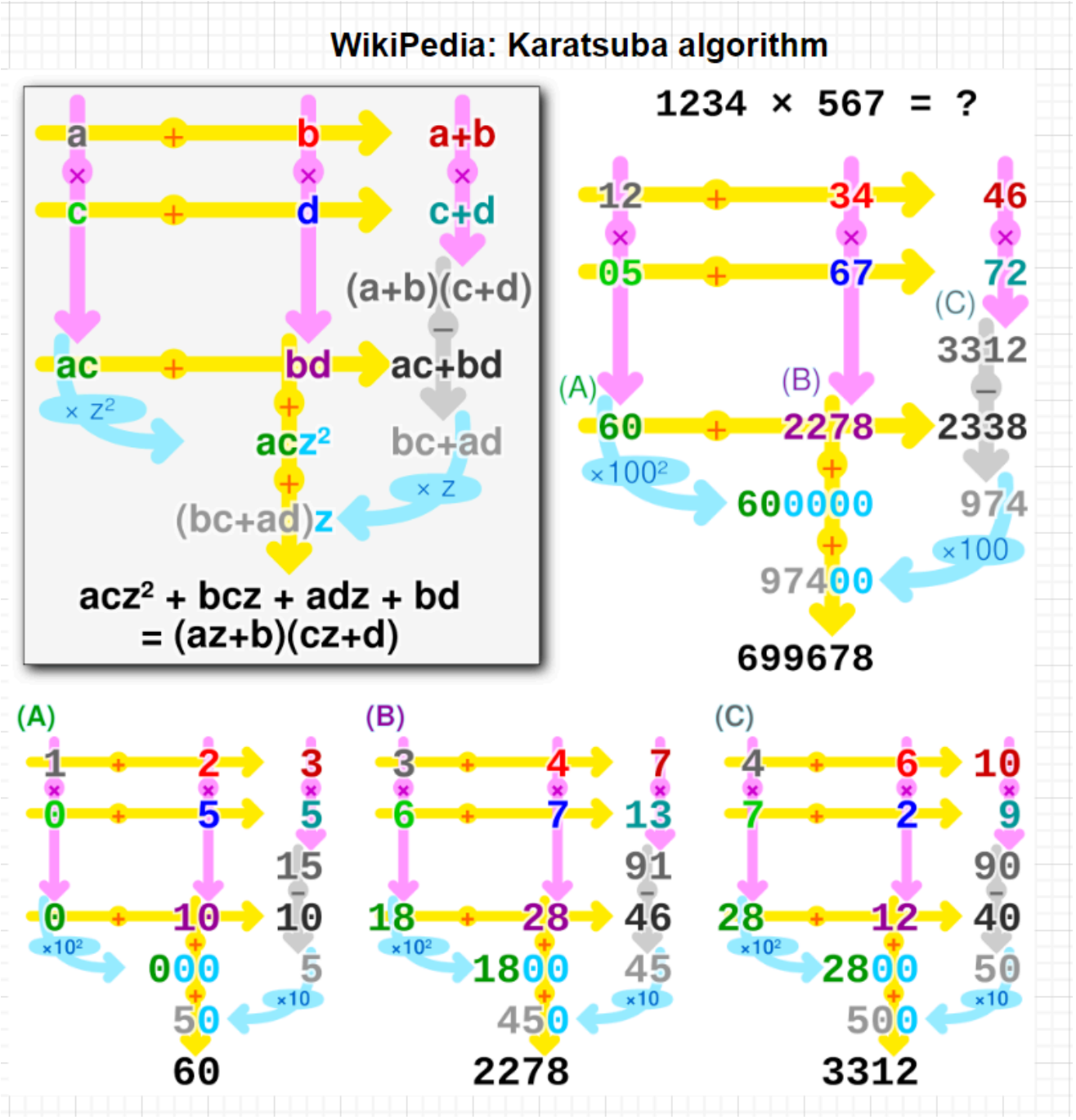


1.3 Methods

- NTT



- Karatsuba



1.4 Parameter defines

- N = 65536
  - degrees of polynomials
  - n = 16
  - $O(2^{16})$
- K = 32
  - data width of coeffitients and modular number
  - (bit)
- q = 998244353
  - modular number

- phi = 629671588
  - `pow(phi, n, q) == 1`
- phi_inv = 283043518
  - `phi * phi_inv % q == 1`
- psi = 24514907
  - `pow(psi, n, q) == -1`
  - `pow(psi, 2*n, q) == 1`
  - `pow(psi, 2, q) == phi`
- psi_inv = 3707709
  - `psi * psi_inv % q == 1`
- N_inv = 998229121
  - `N * N_inv % q == 1`
- inv_2 = 499122177
  - `2 * inv_2 % q == 1`

2 Computation Resources Evaluation

2.1 NTT/INTT (m stages)(1 group)

- Modular Multiplication: $m \cdot \frac{N}{2}$
- Modular Add/Sub: $m \cdot N$

2.2 Iterative Karatsuba (m stages)(n-m groups)

- Modular Add: $2^{n-m} \cdot \sum_{k=0}^{m-1} 2^{m-1} \cdot (\frac{3}{2})^k$
  - summation of geometric progression: $2^{n-m} \cdot 2^m \cdot (\frac{3}{2}^m - 1)$

2.3 Iterative Inv_Karatsuba (m stages)(n-m groups)

- Modular Add/Sub: $2^{n-m} \cdot \sum_{k=0}^{m-1} (2 \cdot (2^{k+1} - 1) + 2 \cdot (2^k - 1))$
  - summation of geometric progression: $2^{n-m} \cdot (6 \cdot (2^m - 1) - 4m)$

2.4 Scheme-1: 2*NTT + odot + INTT

- Modular Multiplication: $3m \cdot \frac{N}{2} + N$
- Modular Add/Sub: $3m \cdot N$

2.5 Scheme-2: 2*Kara + odot + Inv_Kara

- Multiplication: $2^{n-m} \cdot 3^m$
- Modular Add/Sub: $2^{n-m} \cdot (2 \cdot 2^m \cdot (\frac{3}{2}^m - 1) + 6 \cdot (2^m - 1) - 4m)$

2.6 MM vs MA

- Taking K=32bit as an example:
  - Modular Add/Sub: O(3)
  - Multiplication: O(32)
  - Modular Multiplication: O(100)